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A hybrid model of Internet of Things and cloud computing to manage big data in health services applications

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HIGHLIGHTS

- A proposal for improving the health services in industry 4.0 big data applications.
- A new model is proposed to optimize virtual machines selection in cloud-IoT health services applications.
- Three optimizers are used to implement the proposed model.
- The real-time data retrieval is improved by 5.2%.

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ABSTRACT

Over the last decade, there has been an increasing interest in big data research, especially for health services applications. The adoption of the cloud computing and the Internet of Things (IoT) paradigm in the healthcare field can bring several opportunities to medical IT, and experts believe that it can significantly improve healthcare services and contribute to its continuous and systematic innovation in a big data environment such as Industry 4.0 applications. However, the required resources to manage such data in a cloud-IoT environment are still a big challenge. Accordingly, this paper proposes a new model to optimize virtual machines selection (VMs) in cloud-IoT health services applications to efficiently manage a big amount of data in integrated industry 4.0. Industry 4.0 applications require to process and analyze big data, which come from different sources such as sensor data, without human intervention. The proposed model aims to enhance the performance of the healthcare systems by reducing the stakeholders' request execution time, optimizing the required storage of patients' big data and providing a real-time data retrieval mechanism for those applications. The architecture of the proposed hybrid cloud-IoT consists of four main components: stakeholders' devices, stakeholders' requests (tasks), cloud broker and network administrator. To optimize the VMs selection, three different well-known optimizers (Genetic Algorithm (GA), Particle swarm optimizer (PSO) and Parallel Particle swarm optimization (PPSO) are used to build the proposed model. To calculate the execution time of stakeholders' requests, the proposed fitness function is a composition of three important criteria which are CPU utilization, turn-around time and waiting time. A set of experiments were conducted to provide a comparative study between those three optimizers regarding the execution time, the data processing speed, and the system efficiency. The proposed model is tested against the state-of-the-art method to evaluate its effectiveness. The results show that the proposed model outperforms on the state-of-the-art models in total execution time the rate of 50%. Also, the system efficiency regarding real-time data retrieve is significantly improved by 5.2%.

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1. Introduction

Cloud computing offers virtualized computing, storage, and networking resources over the Internet to the organizations and

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individual users in a completely dynamic way [1]. The evolution of Cloud computing is currently related to the increasing popularity of big data which leads to new trends such as industry 4.0 big data analysis. In fact, cloud computing provides the necessary computation, storage, applications, and networking, which can support big data applications [2]. These applications empowered by cloud computing solutions can extract beneficial information to guide better decisions in many usage areas as in healthcare applications [3]. However, healthcare services always pose challenges since different diseases can be developed from time to time. For example, the IoT has been widely applied recently to interconnect available medical resources and provide reliable, efficient and smart healthcare services to the patients with chronic illnesses [4]. The integration of Cloud IoT with industry 4.0 in healthcare systems and applications is representing a combination of communication technologies, interconnected apps, things (devices, sensors, etc.), cloud, and people that would function together as one smart system to monitor, track, and store patients records for ongoing care and analysis in big data environments. Recently, sensors medical devices are installed with the human body to collect precious information about the patients. These sensors are creating an extensive data continuously often called as Big Data. As a result, industry 4.0 applications gained excellent research attention due to their need to analyze big data (sensor data) without human intervention in any of its functions.

Cloud computing simply means that an application is running on computer processors located remotely rather than on a user's on-site computers [5]. Remote storage of data can also be cloud-based. The capacity of the cloud to store and process data is virtually unlimited. Industry 4.0 aims to store and process data remotely which is more economical, flexible and secure than on-site alternatives. The cloud is also more readily scalable, that is, its capacity can be expanded rapidly to meet growing demand in industry 4.0 applications. The connectivity envisioned by the Industrial IoT implies that streams of data will flow to and from connected systems. These streams of data may be massive. Big Data has become the term for the ability to find significant trends or emerging patterns detectable only by rapidly scanning many millions of unstructured data items in different formats from a multitude of diverse sources. The ability to gather and access the enormous mass of data generated by the connected factory (and then make sense of this data) is what Big Data brings to the Industrial IoT.

The integration of Cloud Computing and IoT provides a new storage, processing, scalability and networking capabilities which are so far limited in the IoT due to its characteristics in the area of healthcare [6–9]. Furthermore, new opportunities like artificial intelligence; data mining algorithms (e.g., such as artificial neural network (ANN), ant colony optimization (ACO), genetic algorithms (GA), and simulated annealing (SA), can be applied to analyze big data and mine knowledge, and real-time processing has been presented to be proposed in the healthcare field. Cloud computing transmits application, infrastructure services to big numbers of stakeholders with varied and dynamically changing requirements [1]. Cloud is composed of data centers, hosts, VMs, resources, etc. Datacentres are containing a significant number of resources and list of different applications. Hosts are composed of a large number of VMs to store and regain several medical resources to stakeholders. Cloud computing uses the virtualization technique which permits to share a single physical instance of a resource or an application among various stakeholders and enterprises [2]. It does this by allocating a logical name to physical storage and providing a pointer to that physical resource when requested [10]. The critical challenges of healthcare services are big patients' data, big resources, and big applications by retrieving and storage those processes in the shortest possible time.

Big data has the potential to support many medicinal and healthcare operations, including disease control and management patients' health [11]. The big data is a large amount of stored/acquired data due to advances in various technologies including cloud computing and wireless communication technologies [12]. It is identified by the magnitude of data (volume), kinds of data based on the generation source (variety), and the speed of data generation (velocity) [13]. Some of the big amounts of data could be addressed offline, but some medical applications need real-time processing for this data. Therefore most of the stakeholders prefer to use cloud computing in medical application to save many patients from death [14]. However, handling a huge amount of IoT data in industry 4.0 environment using a cloud computing tool becomes a big problem because of the required execution time and the resources utilization [15–20]. Many reasons that lead to a time delay of medical requests such as waiting time, turnaround time of medical requests and the waste of resources utilization can be avoided if an efficient mechanism is used to manage these data in a cloud IoT environment. One of these mechanisms is proposed in this paper to optimize the virtual machine selection (VMs) in cloud-IoT health services applications to manage patients' big data, reduce time delay and maximize resources utilization on cloud computing environment.

The industry 4.0 is used in many applications such as big data management, smart product, smart city, etc. Big data management works based on large data analysis in a precise and fast way through the use of integrated industry 4.0. Sensors and microchips support the smart product, products in Industry 4.0 become smart. Smart City is a city that comprises six factors in its development policy: smart economy, smart mobility, smart environment, smart people, smart living, and smart governance. By combining the Internet, a telecommunications network, a broadcast network, a wireless broadband network, and sensor networks, IoT will accelerate the development of a new generation IT and knowledge-based economy. Industry 4.0 is used graphs to visualize large amounts of complex data is more accessible than poring over spreadsheets or reports. Visualize large quantities of complex medical data has an essential role in industry 4.0 to help for visualizing medical data accurately.

In the traditional cloud computing healthcare applications, there are a limited number of tasks that need a few counts of VMs to be executed. In that case, it is possible to assign a request to the nearest processing device [21]. However, in complex environments such as most of the recent healthcare applications, it is impossible to do that. Accordingly, parallel processing is the most suitable solution to reduce the execution time of the received requests and maximize resources utilization. IoT is currently used to collect large data from patients efficiently without errors [22]. The proposed model gains the advantages of the parallel processing in cloud environments and the industry 4.0 applications in IoT for big data analysis. This proposed model is implemented using three different well-known optimizers Genetic Algorithm (GA), Particle swarm optimizer (PSO) and Parallel Particle swarm optimizer (PPSO). To calculate the execution time of stakeholders' requests, the proposed fitness function is a composition of three important criteria which are CPU utilization, turn-around time and waiting time. A comparative study between these three optimizers is provided to evaluate the performance of each of them regarding the execution time and the amount of processed data of the medical requests. The proposed model is tested against the state-of-the-art method to evaluate its efficiency. As shown at the related work section, most of the researchers used GA to optimize VMs on the traditional cloud environments; however, we proposed our GA based model on the CloudSim [23] to reduce total execution time of medical requests and maximize utilization of resources. Besides, we introduced the PPSO based model to

optimize the VMs in both traditional and modern cloud computing based environments which are dramatically outperforming the GA results. PPSO outperforms many optimization algorithms such as whale optimizer [24] and GA due to its ability to work in a parallel processing environment. As a result, it saves time, cost and allowing the implementation of applications in shorter wall-clock time.

The rest of this paper is arranged as follows: Section 2 discusses the related works of applying cloud computing and IoT in health services applications, Section 3 describes the proposed hybrid model of cloud computing and IoT for health services, Section 4 explains in details the proposed optimizer using GA, PSO, and PPSO, Section 5 discusses the experimental results and finally, Section 6 concludes the paper.

2. Related work

Through related work, many studies were done on applying and using intelligent algorithms to enhance scheduling and performance of VMs in a cloud-IoT environment such as PSO, ACO, Parallel ACO, and GA as follow:

Gaochao Xu and et al. [25], introduced a new strategy to detect the optimal VMs placement in cloud environment based on distributed parallel ant colony optimization (DPACO). This study explains the importance to find VMs placement in datacentre to maximize performance of VMs, reduce energy cost and maximize resources utilization. Although DPACO gets the optimal VMs placements, it needs some improvements regarding its energy cost.

Y. Tohidirad and et al. [26], proposed a method to implement scheduling of VMs on cloud environment based on first come first served (FCFS) and round robin (RR). This study tries to find the best schedule of VMs in datacenter to reduce energy consumption in servers. The main contribution of this study is to see the best scheduling of VMs in cloud computing environment by using FCFS and RR. It assumes that the optimum selection of the VMs is already known.

M. R. Chowdhury and et al. [27], presented multiple redesigned VMs placement in cloud environment to achieve to save power consumption in servers, maximize resources utilization and improve performance of VMs in datacenter. The clustering of VMs in datacenter based on CPU utilization and RAM allocation helps to enhance performance of VMs to reduce power consumption.

S. E. Dashti and et al. [28], suggested a new method to obtain the optimal VM placement based on modified PSO algorithm. This study tries to find the best VM placement to give quality of services of users' tasks and minimize energy consumption. This proposed method can find the optimal VM placement by modified PSO algorithm to save power consumption and facilitate task scheduling.

V. Ashwin and et al. [29], proposed a new approach to obtain the best VM placement in datacenter based on multi-objective GA and ACO. This study seeks to find optimal VM placement to save power consumption and maximize utilization of resources. This proposed approach can find the optimal VM placement in cloud computing environment by using GA and ACO.

L. Hong and et al. [30], offered a new method to detect optimal scheduling of VMs in datacenter based on genetic ACO algorithm. This study tries to solve VM placement problem to save power consumption, maximize resources utilization and improve performance of VMs in datacenter. The contribution of this study is to detect the best scheduling of VMs in datacenter in cloud environment by genetic ACO.

Abhijitsinh T. Parmar and et al. [31], presented an approach to find optimal VM allocation in cloud environment based on FCFS algorithm. This study tries to find optimal VM allocation to reduce energy consumption, reduce time from users' tasks and facilitate task scheduling. FCFS algorithm may discover optimal VM allocation in cloud computing environment. But, it needs to improve task scheduling.

M. Seddigh and et al. [32], presented a new model to predict scheduling of VMs in datacentre based on ACO algorithm. This study tries to predict of VMs in datacentre to save power consumption. Simulation results are implemented on CloudSim package tool. The proposed model attempts to predict scheduling of VMs in cloud environment by using ACO.

Y. Wang and et al. [33], presented a new method to find optimal VM placement based on mixed integer programming formulation. This study tries to find optimal VM placement to reduce total power of all servers in datacentre and maximize resources utilization. The proposed method outperforms on standard integer programming algorithms in cloud computing environment and blur of optimizing resources utilization.

T. Thiruvenkadam and et al. [34], presented a new approach to detect the best VM placement in datacentre based on RR and FCFS algorithms. The target of VM placement maximizes resources utilization; reduce response time Power Usage, Load Imbalance Rate, Migration Rate and better services to the cloud users. The proposed approach may detect the best VM placement in cloud computing environment by RR and FCFS. However, it needs to enhance the response time of requests.

E. Barlaskar and et al. [35], proposed a new model to find energy efficient of VM placement based on the hierarchical cluster based modified firefly algorithm (HCMFF). The goal of energy VM placement is to optimize energy consumption. The simulation results show that HCMFF outperforms on honeybee algorithm and original Firefly. The proposed model can reduce energy VM placement in cloud environment by HCMFF.

Md. Mofijul Islam and et al. [36], suggested a novel approach to finding VM migration (VMM) on mobile cloud computing based on GA. The target of GAVMM is to see the optimal cloud host for a mobile VM and to reduce the total number of VM migrations, resulting in a minimized task implementation time. The contribution of this study is to find VM migrations in datacentre in cloud environment by GAVMM.

A. Mosa and et al. [37], presented a new method to select optimal VM placement in datacentre based on GA. This study tries to choose optimal VM placement to reduce power consumption in servers, reduce response time and maximize resources utilization. The proposed method works to select optimal VM placement in cloud environment by using GA.

R. S. Moorthy and et al. [38], presented a new mechanism to find VM placement in datacentre based on ACO. The proposed mechanism focuses on reducing the cost spending in each plan for hosting VMs in a multiple cloud provider and the response time of each cloud provider is monitored periodically, in such a way to reduce delay in providing the resources to the users. ACO may reduce the cost, response time and failed to reduce the number of migrations.

A. Shrivastava and et al. [39], presented a new method to find VM allocation in datacentre based on best fit decreasing. This study tries to solve VM allocation in datacentre to improve resources utilization. The proposed method may solve VM allocation problem in cloud computing environment and failed to reduce total execution time of requests.

S. Rallapallia and et al. [40], introduced a novel framework to process and analyze big scale healthcare data on cloud environment based on Hadoop. This study explains the importance of examining large data to take a right decision to patients for choosing the proper care. The proposed framework can analyze large data of healthcare appropriately by using Hadoop.

J. Hanen and et al. [41], introduced a new system to help patients to treat heart disease remotely based on medical cloud multi-agent system (MCMAS). MCMAS is applied to Google's Android operating system and CloudSim for solution traditional problem in diagnosis heart disease. MCMAS may find the better solution for diagnosis heart disease on mobile cloud computing.

A. ALEXANDRU and et al. [42], introduced a survey in big data on cloud computing in healthcare services. This study shows different tools to store and analyze patients' data on cloud environment. It proposes big data architecture for healthcare. There are no details about the proposed structure and its tools used for big data analyze.

L. A. Tawalbeh and et al. [43], introduced a new model to analyze big patients data real-time by mobile cloud computing.). A new model is applied to Google's Android operating system. The proposed model can analyze big patients data by using mobile cloud computing.

G. Manogaran and et al. [44], introduced a new architecture based on big data and industry 4.0 to present a smart solution to secure data in healthcare. The proposed system utilizes critical management security technicality to keep big data in industry 4.0. The proposed model can be considered as a new smart architecture to secure data in healthcare applications.

Y. MA and et al. [45], introduced a new approach to present intelligent solution to uneven process distribution of medical resources (UDMR), the growing chronic diseases (GCD) and the increasing medical expenses (IME). This study presents the big health application system based on the health Internet of Things and big data.

C. Thuemmler and et al. [46], introduced a new model to support the pharmaceutical industry based on IOT and industry 4.0. The new model is based on smartphones technology and smart applications to support the pharmaceutical industry as an example of healthcare.

D. Dimitrov [47] introduced a new framework to support health education in a cloud environment based on IoT and big data. Smart sensors are used to collect big data from patients to help health education for a set of rare diseases. However, additional details and experimental results about the proposed model is required to evaluate its performance, especially in complex applications.

R. Liand et al. [48], presented a new method to detect a VM placement in datacentre based on Multi-objective optimization. The proposed method focused on reducing the cost spending in each plan for hosting VMs in a cloud provider. The response time of each cloud provider is monitored periodically, in such a way to reduce delay in providing the resources to the users. Although the multi-objective optimization can reduce the response time and the number of migrations, it needs a high processing capability.

Therefore, our proposed model can be differentiated from the above state-of-the-art models by offering a PPSO parallel processing algorithm for IoT big data analysis in a cloud computing healthcare applications. It is proposed for the applications with large-scale resources and tasks. However, our proposed model is not recommended for the applications that need processors overuse to execute the functions on a cloud computing environment.

3. The proposed intelligent model of cloud computing for HCS

This section describes the architecture of the proposed cloud-IoT in integrated industry 4.0 for HCS. It consists of four main components: stakeholders' devices, stakeholders' requests (tasks), cloud broker and network administrator as shown in Fig. 1. The communication devices services are responsible for implementing different network communication management between stakeholders and the cloud.

Stakeholders use a variety of IoT devices (PC, Laptop, Smartphone, Tablet, Digital sensor and etc.) to send a variety of medical requests (tasks) efficiently through cloud computing to obtain different preventive services such as retrieving a patient's data, telemedicine, diagnosis of diseases, electronic medical records (EMR) and etc. IOT endpoints are delivered by big medical data to clouds through the internet to configure a substantial medical database to improve medical services. Cloud broker acts as a mediator who is responsible for sending and receiving requests (tasks) over the cloud. Each network may have several application hosts = {Host1, Host2,…, and Hostn} providing the SaaS and can be allocated to execute the cloud stakeholder's requests. Each application host has a set of resources = {R1, R2 …and Rn} that can be assigned for the coming stakeholders' requests. Each network has a network administrator that is responsible for the coordination of the communication between the hosts inside the network and between this network and other networks in the clouds. Network administrator is responsible for applying the most efficient intelligent techniques (e.g. PPSO algorithm) that it uses to find the optimal selection of VMs in cloud to enhance the task scheduling process which leads to reducing total execution time of stakeholders' requests (tasks), Exploiting the advantage of the speed of time in the transfer of medical services by integrated industry 4.0 and maximize utilization of resources.

4. The proposed GA, PSO and PPSO based algorithms for cloud-IoT big data model

The proposed architecture is executed using three different algorithms, GA, PSO and PPSO. The proposed fitness function is composed of three essential attributes which are CPU utilization, turn-around time and waiting time. The calculations of these attributes depend on the following three parameters:

- **Arrival Time (AT)**: the time at which the task arrives in the ready queue.
- **Burst Time (BT)**: the time required by a task for CPU execution.
- **Completion Time (CT)**: the time at which the task completes its execution.

This section presents the formulas for calculating the three important attributes using the formerly aforesaid parameters to find the optimal selection of VMs, as follows:

$$\text{CPU Utilization (U)} = 100\% - (\%\text{time spent in the idle task}) \quad (1)$$

$$\text{Turnaround Time (TT)} = TT = CT - AT \quad (2)$$

$$\text{Waiting Time (WT)} = WT = TT - BT. \quad (3)$$

In the following subsections, the three different algorithms that are used to implement our proposed cloud-IoT in integrated industry 4.0 for HCS in a big data application are described as shown below in algorithm 1 and algorithm 2. The proposed algorithms try to find the optimal selection of VMs to reduce the execution time of requests (tasks) from stakeholders and maximization of resources utilization.

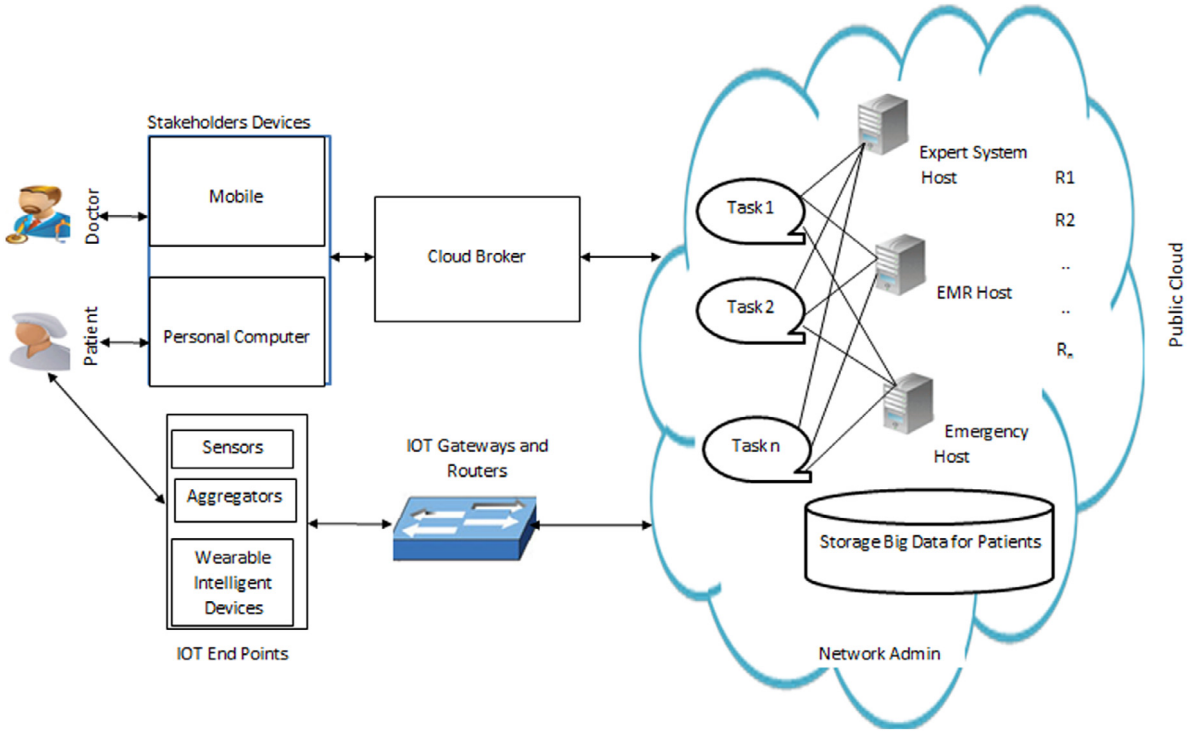


Fig. 1. Intelligent model of cloud-IOT for HCS.

4.1. The proposed cloud-IoT model using GA

In this subsection, we extensively discuss the proposed GA algorithm that is used to implement our proposed Cloud-IoT model by integrated industry 4.0 concept. The importance of population size, crossover and mutation probabilities are generating new offspring that helps to find new solutions [49].

Assume that there are M GA population $_{\text{chromosomes}}(\text{VMs}) = 100$ and number of iteration $= 10$. Each VM in the cloud(s) is considered a chromosome which represents a potential solution (VM) that can be allocated for executing the stakeholder's subtasks. Compute fitness function (optimal selection of VMs) by using U, TT, and WT. If stakeholder's task is finished execution, find the optimal execution time and optimal choice of VMs, otherwise, implement selection operation, crossover operation, mutation operation, elitism method and generate new population, as shown below:

Selection operation helps to keep the best chromosomes (VMs) in the population and selecting a fitting pair from each chromosome (VMs).

The proposed model is based on the cut two points (two VMs on the cloud) randomly and compares them to generate the best one [50]. So, implement the two points crossover operation between two chromosomes (two VMs) and obtain two different offspring. Crossover is calculated by using two equations as follows:

$$\text{Offspring1} = A * \text{Parent 1} + (1 - A) * \text{Parent 2} \tag{4}$$

$$\text{Offspring 2} = (1 - A) * \text{Parent 1} + A * \text{Parent 2} \tag{5}$$

where: A is represented a random number (selected before each crossover operation), and parent is VM.

Mutation method aims to hold simple changes in chromosomes [51]. So, the proposed model uses flip bit mutation method for that purpose. Mutation updates one or more gene values in a chromosome (VM) from its initial state. Mutation $= 0.6$.

Elitism operation of the chromosomes (VMs) means to store the optimal chromosomes into the system for future use. GA on cloud computing can be applied as in Algorithm 1:

The GA aims to calculate the fitness function (optimal selection of VMs) by using U, TT and WT for each chromosome (VM) to select the optimal selection of VMs through the smallest execution time for each chromosome (VM) on a cloud environment.

4.2. The proposed cloud-IoT model using PSO

Assume that there are M particles (VMs) $= 100$, $C_1 = 2$, $C_2 = 2$ and the number of iterations $= 10$. Inertia weight is an important parameter in PSO, which significantly affects the convergence and exploration–exploitation trade-off in PSO process. Each VM in the cloud(s) is considered a particle which represents a potential solution (VM) that can be allocated for executing the stakeholder's subtasks. Compute fitness function (optimal selection of VMs) by using U, TT, and WT. Compare the calculated fitness function of each particle (VM) with its pbest_i. If the current value is better than pbest, then put the current location as pbest location. Moreover, if the current value is better than gbest, then reconstruct gbest to the current index in particle array. Assign the best particle (VM) as gbest. Update each Particle Velocity and position according to Eqs. (6), (7) as follows:

The velocity value is computed by Eq. (6) [52].

$$V_i(t+1) = V_i(t) + U_1C_1 \times (P_{\text{p-best}} - X_i(t)) + U_2C_2 \times (P_{\text{g-best}} - X_i(t)) \tag{6}$$

where:

- $V_i\,(t+1)$ represents the new velocity of a particle and $V_i\,(t)$ represents its current velocity.
- U_1 and U_2 are two random variables in the range [0, 1].
- The constants C_1 and C_2 represent the learning factors.
- The x-vector records the current position of the particle in the search space.

Algorithm 1. The Proposed GA for Selecting VMs on Cloud Computing

```
Input : Size α of Population,
        Rate β of Elitism,
        Rate Y of Mutation,
        Number σ of Iterations
Output : X ⟵(Optimal Chromosomes (VMs), Optimal Exec Time)
Stakeholders Tasks = ST;
Count I = 0;
Generate a random σ solutions;
While (ST ≠ 0)
For I = 0 to σ do
   Compute fitness function;
   X = fitness function;
If (ST is finished == true)
     ST= ST – 1;
     Save X;
Else
Elitism Operation(EO, POP, E);
Two Points Crossover Operation(S, D, F);
Bit Flip Mutation Operation();
Update K = K1 + K2;
End if
I = I + 1;
End for
End while
Return the best solution (VM) X in K;
Elitism Operation(ne, α, β );
Begin
   Number of elitism   ne = α. β;
   Select the best ne solutions (VMs) in α and save them in K1;
   Return K1;
End
Crossover Operation(nc, α, ne );
Begin
   Number of crossover  nc = (α - ne) / 2;
For j = 1 to nc do
   Randomly select two solutions (two VMs) XA and XB from α;
   Generate XC and XD by one point crossover to XA and XB;
   Save XC and XD to K2;
   Return K2;
End
End for
Mutation Operation();
Begin
For j = 1 to nc do
    Select a solution (VM) XJ from K2;
    Mutate each bit of XJ under the rate Y and generate a new solution XL;
If XL is useless
  Update XL with a feasible solution by repairing XL;
End if
    Update XJ with XL in K2;
    Return K2;
End
End for
```

- $P_{\text{p-best}}$ is the best particle agent I.
- $P_{\text{g-best}}$ is the best particle in search space.

The position value is computed by Eq. (7) [53].

$$X_i(t+1) = X_i(t) + V_I(t+1) \tag{7}$$

where

- $X_i\,(t+1)$ represents the new position of a particle
- $X_i(t)$ represents its current position.

If a stakeholder's task is finished, the algorithm finds the optimal execution time and the optimal selection of VMs; otherwise, it calculates the fitness function (optimal selection of VMs) for each particle (VM).

Similar to GA, the PSO algorithm aims to calculate the fitness function (optimal selection of VMs) by using U, TT and WT for each chromosome (VM) to select the optimal selection of VMs through the smallest execution time for each chromosome (VM) on a cloud environment as shown at Algorithm 2.

4.3. The proposed cloud-IoT model using PPSO

The third proposed algorithm for the could-IoT model is based on PPSO. In PPSO, the parallel processing aims to produce the same results of the classical PSO using multiple processors simultaneously with the goal of reducing the runtime. The same steps described in PSO will be applied with some updates as listed below:

Algorithm 2. The Proposed PSO for Selecting VMs on Cloud Computing

```
Input : Size α of Population,
        Rate β of P_g-best,
        Rate σ of P_p-best,
        Number Y of Iterations
Output : P_g-best <- (Optimal Particles (VMs), Optimal Exec Time)
Stakeholders Requests = X;
Count I = 0;
For I = 0 to α do
    P_velocity <- Random velocity();
    P_position <- Random position(α);
    σ <- P_position;
If (cost(σ) <= cost(β))
    β <- σ;
  End if
End for
While (X ≠ 0)
For I = 0 to Y do
  Calculate fitness function;
  Update velocity (P_velocity, β, σ);
  Update position (P_velocity, P_position);
    P_velocity <- Update velocity;
    P_position <- Update position;
If (cost(P_position) <= cost(σ))
    σ <- P_position;
If (cost(σ) <= cost(β))
    β <- σ;
If (X is finished == true)
     X= X – 1;
     Save β;
Else
     Compute fitness function for each
     particle(VM);
     End if
   End if
End if
I = I + 1;
   End for
End while
Return β;
```

(a) PPSO will define how many groups of processors are needed for the fitness function (optimal priority scheduling of VMs) to be executed because it can be designed to be $2n$ sets.

(b) The barrier synchronization stops the algorithm from the move to the next step until the fitness function (optimal selection of VMs) has been reported, which is required to maintain algorithm coherence. Ensure that all of the particle (VM) fitness evaluations have been completed and results reported before the velocity and position calculations can be executed.

(c) Update particle velocity according to Eq. (8) [54].

$$\begin{aligned} V_{l,j}(t) &= WV_{l,j}(t-1) \\ &+ C_1R_1(P_{i,j}(t-1) - X_{i,j}(t-1)) \\ &+ C_2R_2(P_{s,j}(t-1) - X_{i,j}(t-1)) \\ &+ C_3R_3(P_{g,j}(t-1) - X_{i,j}(t-1)) \end{aligned} \tag{8}$$

where:

- $X_{i,j}$ = the position of ith particle in jth swarm,
- $V_{l,j}$ = the velocity of ith particle in jth swarm,
- $P_{i,j}$ = the pbest of ith particle in jth swarm,
- $P_{s,j}$ = the swarm best of jth swarm,
- Pg, j = the global best among all the sub swarms,
- W = inertia weight,
- C_1, C_2, C_3 = acceleration parameters
- R_1, R_2, R_3 = the random variables.

(d) Update particle position according to Eq. (9).

$$X_{i,j}(t) = X_{i,j}(t-1) + V_{l,j}(t) \tag{9}$$

$X_{i,j}(t)$ = the current position of ith particle in jth swarm, $X_{i,j}(t-1)$ = the new position of ith particle in jth swarm, $V_{l,j}(t)$ = the current velocity of ith particle in jth swarm.

5. Experimental results

This section discusses the experimental results of our proposed cloud-IoT model with industry 4.0 using the three proposed algorithms GA, PSO and PPSO. These three algorithms are implemented using two different tools to ensure their consistency. The first tool is MATLAB while the second one is the CloudSim package.

A comparison between the results of these three algorithms is conducted to evaluate their performance regarding the execution time, the speed of live data processing and the system efficiency.

The experiments are conducted based on a proposed performance metrics which are composed of speedup and efficiency. The Speedup is the gauge used to calculate the proportion of the average implementation time on one processor $A[t_1]$ and the average implementation times on multiple processors $A[t_k]$. The speedup due to k processors is clarified by [54]:

$$S_K = A[t_1] \div A[t_k]. \tag{10}$$

Suppose the k value is 8. The implementation times of GA, PSO and PPSO are shown in Tables 2, 4 and 6. The speedup of PPSO is 3.51 times faster than GA. The speedup of PPSO is 1.42 times faster than PSO. The Efficiency is clarified as the percentage of speedup

Table 1
GA parameters.

No	Parameters	Values
1	Population size	100–1000
2	Crossover probability	0.5
3	Crossover type	Two points
4	Mutation probability	0.6
5	Mutation type	Bit flip
6	Number of iterations	10

Table 2
GA results.

Number of trials	Populations	Execution time (ET)
1	100	10.24
2	200	11.78
3	300	17.03
4	400	23.70
5	500	30.10

Table 3
PSO parameters.

No	Parameters	Values
1	Number of particles	100–1000
2	C_1	2
3	C_2	2
4	W_max	0.9
5	W_min	0.3
6	Number of iterations	10

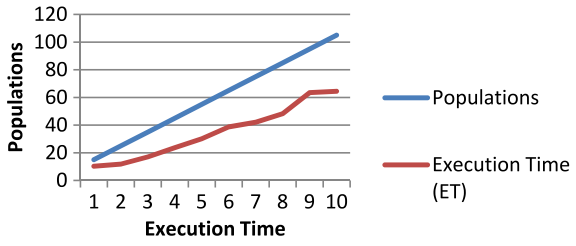


Fig. 2. The relationship between Populations and ET in GA.

by k processors. It is presented by [54]:

$$E_k = (S_K \div K) \times 100\%. \quad (11)$$

The required parameters to run GA are shown in Table 1.

By running GA using the parameters at Table 1, the results at Table 2 are obtained. Table 2 shows a sable of 5 iterations, populations and their corresponding execution time. As known in GA, the execution time increases by increasing the population sizes due to the need to search n a vast amount of possible solution. That is why the value of ET is increased by increasing the number of trials.

Simultaneously, Fig. 2 shows the positive relationship between the number of populations and the corresponding execution time. Due to its randomness, the mutation process causes a random jump in the location of the generated solutions. That can describe the reason of the curved dropping in trials number 7 and 8.

Regarding PSO, it uses the same parameters that are used by GA at equations number (1), (2), (3). Besides, a set of parameters are assigned to it as shown in Table 3.

Using the above PSO parameters, Table 4 shows the obtained results of our proposed model. The ET of PSO is much better than the corresponding ET of GA. In this case, also, the ET increases by increasing the particles in the different trials. This positive relationship is shown in Fig. 3.

By using a set of parallel processors, the performance of PSO can be improved in PPSO. Dependently, the execution time can be minimized using the same parameters that are used by the classical

Table 4
PSO results.

Number of trials	Particles	ET
1	100	3.08
2	200	4.91
3	300	6.82
4	400	8.91
5	500	11.76
6	600	14.92
7	700	18.91
8	800	21.07
9	900	24.80
10	1000	26.02

Table 5
The efficiency of the PPSO algorithm.

Number of trials	P	S_P
1	2	1.5
2	4	2.5
3	8	4.5
4	16	8.5
5	32	16.5
6	64	32.5
7	128	64.5
8	256	128.5
9	512	256.5
10	1024	512.5

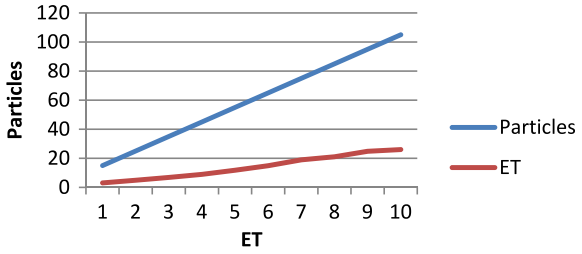


Fig. 3. The relationship between particles and ET in PSO.

PSO as shown in Table 5. The same procedures described in PSO will be implemented, but in PPSO will determine the number of processors needed for the fitness function to be performed. The proposed model can be used in environments with large-scale resources and tasks. For that, Gustafson law is preferred due to its ability to deal with some parallel processing tasks. The efficiency of the PPSO algorithm can be estimated using Gustafson's Law Eq. (12).

$$S_P \leq P - \alpha \times (P - 1) \quad (12)$$

where:

- S_P is speed up
- α is the fraction of non-parallelized stakeholder's requests where the Parallel work per processor is fixed.
- P is the number of processors.

Suppose $\alpha = 0.5$ and number of processors (2, 4, 8, 16, 32, 64, 128, 256, 512, and 1024).

• If $\alpha = 0.5$ and $P = 2$:

$$S2 \leq 2 - 0.5 \times (2 - 1) = 1.5.$$

Increasing the number of processors can dramatically affect the running time. Fig. 4 shows the relationship between the used number of processors and the time speedup.

To test the PPSO efficiency in different environments, an additional experiment is conducted by multiply the data size twice and the results are shown in Table 6. Although the ET increased in this experiment, the variance between these results and the results of the previous experiment is still acceptable with bigger data size.

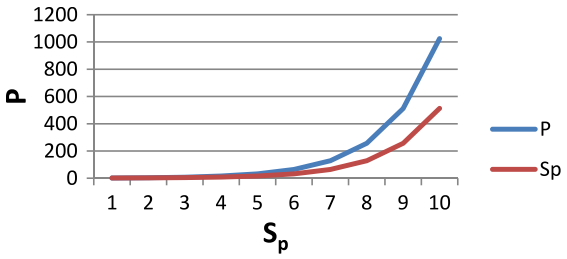


Fig. 4. The relationship between P and S_P in parallel processing.

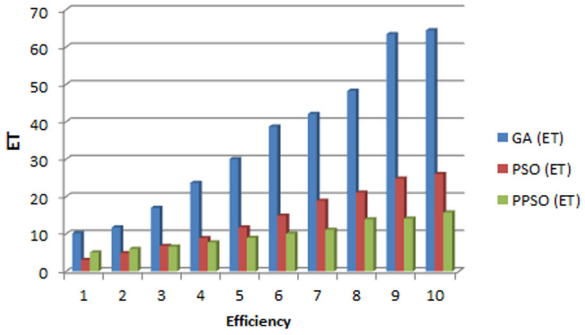


Fig. 5. The relationship between GA, PSO and PPSO.

Table 6
PPSO results.

Number of trials	Particles	ET
1	100	5.05
2	200	6.01
3	300	6.68
4	400	7.77
5	500	8.98
6	600	10.11
7	700	11.14
8	800	13.99
9	900	14.15
10	1000	15.78

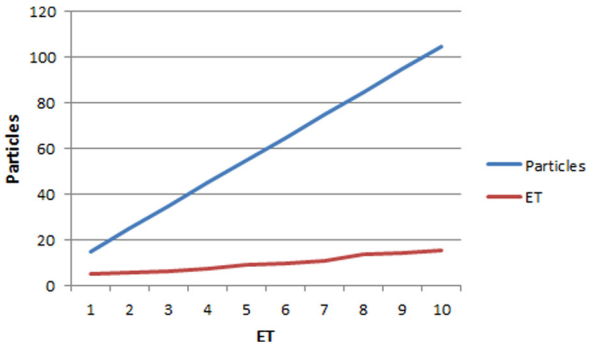


Fig. 6. The relationship between particles and ET in PPSO.

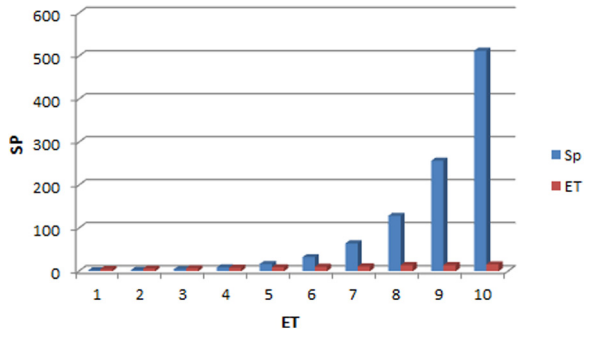


Fig. 7. The relationship between S_P and ET in PPSO.

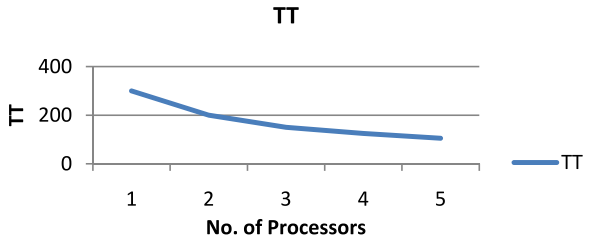


Fig. 8. Proposed PPSO leads to minimized TT with increasing in no. of processors.

From these experiments, we can conclude that PPSO algorithm is 43.9% efficient than GA and 17.8% efficient than PSO. PPSO algorithm is efficient than GA and PSO for stakeholders to reduce the execution time of requests (tasks (T)). Fig. 5 supports this result by providing a comparative view of the three proposed algorithms to evaluate their efficiency regarding ET using the same parameters. As shown in Fig. 8, PPSO outperforms on all of them.

Dependently, Fig. 6 illustrates the change ratio of the ET to reflect its relationship to the number of particles.

Retrieving the real-time data in the shortest time is one of our model's goals. In healthcare applications, real-time data retrieval enables the stakeholders to take the right decision at the right time. To evaluate the performance of the proposed Cloud-IoT model with industry 4.0 regarding the speed of real-time data retrieval, PPSO is used. In this experiment, we tested PPSO only based on its results as discussed before. As shown in Fig. 7, the speed of real-time data retrieval and the ET are increased with the continuous iterations.

To prove the consistency of our proposed PPSO algorithm for the cloud-IoT model by integrated industry 4.0 concept, a set of experiments are conducted on CloudSim to find the optimal selection of VMs. The first experiment uses the default CloudSim where the first task takes the first VM, and the second task takes the second VM, etc. The results show that the total time that is required to build a successful cloudlet is 3 s as shown in Table 7.

The second experiment is conducted by allowing the task x to be assigned VM_x or VM_{x+1}. In this experiment, the total required time to build successful cloudlets is 1 s as shown in Table 8.

Table 9 shows a sample of results of PPSO algorithm on CloudSim regarding turnaround time, waiting time and CPU utilization. Each value represents in Table 9 by the number of trials on CloudSim for example waste of CPU utilization. It decreased based on many reasons such as change the number of processors in each test, change the number of tasks in each trial, etc. Besides, it shows how PPSO suggests the scheduling of tasks on VMs. There is an inverse relationship between the number of processors and TT, WT, and the waste of CPU utilization, as shown below in Figs. 8–10.

Fig. 8 shows the inverse relationship between the number of processors and the turnaround time of requests. Which means, increasing the number of processors reflects the decreasing in the turnaround time.

Simultaneously, Figs. 9, 10, and 11 illustrate the inverse relationship between the number of processors and the waiting time, CPU utilization, and the Makespan (time), respectively.

Regarding the resource utilization, we use Eq. (13) to get its value. Accordingly, Fig. 12 tests the performance of PPSO regarding the resource utilization [55].

$$RUi = MS \times 100/TRU \tag{13}$$

Table 7
Results of default CloudSim.

Cloudlet ID	Status	Datacentre ID	VM ID	Time	Start time	Finish time
0	Success	2	0	800	0.1	800.1
1	Success	2	1	1200	0.1	800.1
3	Success	2	3	8000	0.1	800.1
2	Success	2	2	16000	0.1	800.1
Build successful (total time: 3 s)						

Table 8
A sample of results of PPSO algorithm on CloudSim.

Cloudlet ID	Status	Datacentre ID	VM ID	Time	Start time	Finish time
0	Success	2	1	1600	0.1	1600.1
1	Success	2	1	2000	0.1	2000.1
3	Success	2	3	8000	0.1	8000.1
2	Success	2	2	16000	0.1	16000.1
Build successful (total time: 1 s)						

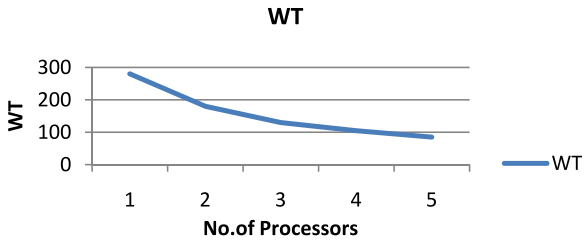


Fig. 9. Proposed PPSO leads to minimized WT with increasing in no. of processors.

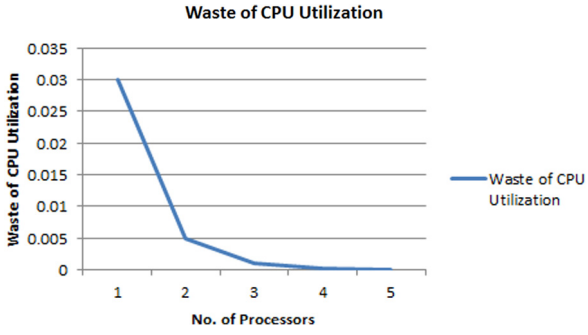


Fig. 10. Proposed PPSO leads to minimized waste of CPU utilization with increasing in no. of processors.

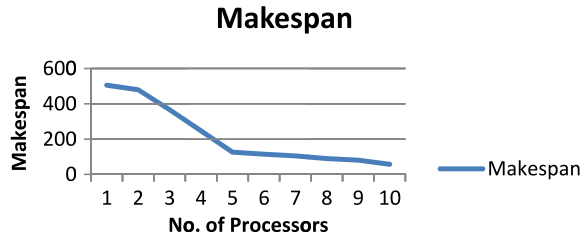


Fig. 11. Proposed PPSO leads to minimized makespan with increasing in no. of processors.

where:

- *RUi* = Usage of Resource in percentage
- MS = Makespan of the task
- TRU = Total Resource Usage—it is the total time for which the Resource *i* is used
  Suppose:
- Total Resource Usage = 500
- Total time for which the Resource *i* is used in VM0 = 879.99

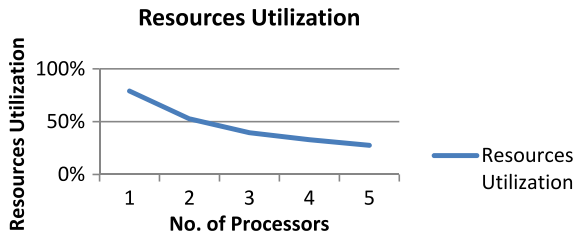


Fig. 12. The relationship between resources utilization and no. of processors.

Table 9
A sample of scheduling of tasks on VMs by using PPSO algorithm on CloudSim.

VMs	Tasks			
	T0	T1	T2	T3
VM1	×	×	–	–
VM2	–	–	×	–
VM3	–	-	–	×
TT	1600	2000	8000	16000
WT	1580	1980	7980	15980
Waste of CPU utilization	0.006%	0.005%	0.0013%	0.00006%

Table 10
A sample of total ET of the intelligent algorithms in the related work and the proposed model.

No	Intelligent algorithms in the related work	Total ET (s)
1	S.E. Dashti and et al. [12]	6.95
2	Abhijitsinh T. Parmar and et al. [15]	7.82
3	R.S. Moorthy and et al. [22]	1.5
4	J. Hanen and et al. [25]	3
5	The proposed PPSO model	1

- Makespan of the task = 299.99

  ∗ *Solution* : $RUi = 299.99 \times 100/379.99 = 79\%$.

Compared to some of the state-of-the-art methods which are mentioned in related work, Table 10 introduces the total execution time of our proposed model. The results show that the improvement rate regarding the second best method is in the range 50% to 88%. This improvement can be observed as shown in Fig. 13.

Fig. 14 shows a comparison between the proposed model and the state-of-the-art methods regarding the system efficiency. The results show that the proposed model dramatically improves the system efficiency by 5.2%.

The proposed model has a high flexibility to change the number of processors, tasks, VMs, etc. quickly. The robust of the proposed model is clear concerning the efficiency of PPSO that outperformed

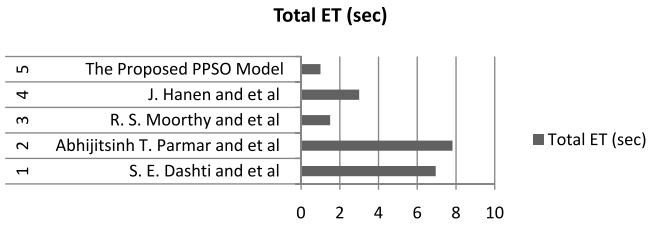


Fig. 13. The total ET of the proposed model compared to the state-of-the-art models.

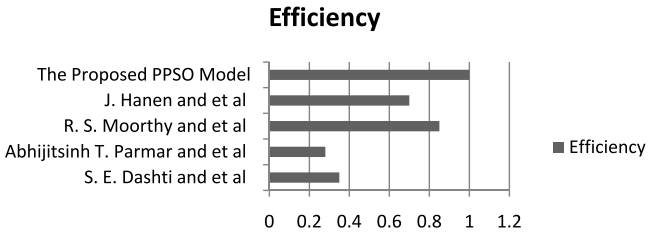


Fig. 14. The system efficiency of the proposed model compared to the state-of-the-art models.

the ability of PSO, GA and the state-of-the-art techniques and it succeeded to select the optimal VMs in a cloud-IoT environment. The versatile of the proposed model can adapt smoothly from one to another of various tasks.

6. Conclusion

Cloud Computing and IoT integration provide a new storage, processing, scalability and networking capabilities which are so far limited in the IoT due to its characteristics in the area of healthcare. However, big data processing in complex environments such as industry 4.0 applications is still an open challenge. To solve this problem, this paper proposes a new model for Cloud-IoT based health service applications in integrated industry 4.0 environment. The proposed model aims to improve the health services performed in a cloud-IoT and industry 4.0 based environment by optimizing the selection of VMs. The proposed model is implemented using three different proposed algorithms using GA, PSO, and PPSO optimizers. The proposed model aims to find the best selection of VMs to help stakeholders in reducing execution time, turnaround time and waiting time of medical requests (tasks), enhancing task scheduling, maximizing utilization of resources, and accessing storage patient's data efficiently. To calculate the execution time of stakeholders' requests, the proposed fitness function is a composition of three important criteria which are CPU utilization, turn-around time and waiting time. A set of experiments were conducted to provide a comparative study between those three optimizers regarding the execution time, the data processing speed, and the system efficiency. The proposed model is tested against the state-of-the-art method to evaluate its effectiveness. The results show that the proposed model outperforms on the state-of-the-art models in total execution time the rate of 50%. Also, the system efficiency regarding real-time data retrieve is significantly improved by 5.2%.

Future research work will focus on the task scheduling process. Additional experiments will be conducted to evaluate the proposed model performance in different environments, especially the applications with a limited count of processors which need processor overuse.

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